

3. Plants are more readily manipulated by genetic engineering than are animals because
 - (A) plant genes do not contain introns.
 - (B) more vectors are available for transferring recombinant DNA into plant cells.
 - (C) a somatic plant cell can often give rise to a complete plant.
 - (D) plant cells have larger nuclei.
4. A paleontologist has recovered a bit of tissue from the 400-year-old preserved skin of an extinct dodo (a bird). To compare a specific region of the DNA from a sample with DNA from living birds, which of the following would be most useful for increasing the amount of dodo DNA available for testing?
 - (A) SNP analysis
 - (B) polymerase chain reaction (PCR)
 - (C) electroporation
 - (D) gel electrophoresis

Levels 3-4: Applying/Analyzing

5. Which of the following is true of cDNA produced using human brain tissue as the starting material?
 - (A) The procedure to make it requires amplification by the polymerase chain reaction.
 - (B) It is produced from pre-mRNA using reverse transcriptase.
 - (C) It can be labeled and used as a probe to detect genes expressed in the brain.
 - (D) It includes the introns of the pre-mRNA.
6. Expression of a cloned eukaryotic gene in a bacterial cell involves many challenges. The use of mRNA and reverse transcriptase is part of a strategy to solve the problem of
 - (A) post-transcriptional processing.
 - (B) post-translational processing.
 - (C) nucleic acid hybridization.
 - (D) restriction fragment ligation.
7. Which of the following sequences in double-stranded DNA is most likely to be recognized as a cutting site for a restriction enzyme?
 - (A) AAGG
TTCC
 - (B) GGCC
CCGG
 - (C) ACCA
TGGT
 - (D) AAAA
TTTT

Levels 5-6: Evaluating/Creating

8. **MAKE CONNECTIONS** Imagine you want to study one of the human crystallins, proteins present in the lens of the eye (see Figure 1.8). To obtain a sufficient amount of the protein of interest, you decide to clone the gene that codes for it. Assume you know the sequence of this gene. Explain how you would go about this.
9. **MAKE CONNECTIONS** Looking at Figure 20.15, what does it mean for a SNP to be “linked” to a disease-associated allele? How does this allow the SNP to be used as a genetic marker? (See Concept 15.3.)
10. **DRAW IT** You are cloning an aardvark gene, using a bacterial plasmid as a vector. The green diagram shows the plasmid, which contains the restriction site for the enzyme used in Figure 20.5. Above the plasmid is a segment of linear aardvark DNA that was synthesized using PCR. Diagram your cloning procedure, and show what would happen to these two molecules during each step. Use one color for the aardvark

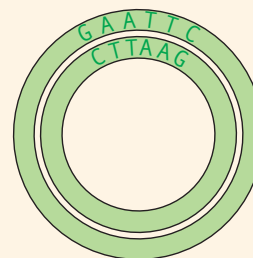
DNA and its bases and another color for those of the plasmid. Label each step and all 5' and 3' ends.

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5' GAATTCTAAAGCGCTTATGAATTC 3'
3' CTTAAGATTTTCGCGAATACTTAAG 5'

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Aardvark DNA



Plasmid

11. **EVOLUTION CONNECTION** Ethical considerations aside, if DNA-based technologies became widely used, discuss how they might change the way evolution proceeds, as compared with the natural evolutionary mechanisms that have operated for the past 4 billion years.
12. **SCIENTIFIC INQUIRY** You hope to study a gene that codes for a neurotransmitter protein produced in human brain cells. You know the amino acid sequence of the protein. Explain how you might (a) identify what genes are expressed in a specific type of brain cell, (b) identify (and isolate) the neurotransmitter gene, (c) produce multiple copies of the gene for study, and (d) produce large quantities of the neurotransmitter for evaluation as a potential medication.
13. **WRITE ABOUT A THEME: INFORMATION** In a short essay (100–150 words), discuss how the genetic basis of life plays a central role in biotechnology.
14. **SYNTHESIZE YOUR KNOWLEDGE**



The water in the Yellowstone National Park hot springs shown here is around 160°F (70°C). Biologists assumed that no species of organisms could live in water above about 130°F (55°C), so they were surprised to find several species of bacteria there, now called *thermophiles* (“heat-lovers”). You’ve

learned in this chapter how an enzyme from one species, *Thermus aquaticus*, made feasible one of the most important DNA-based techniques used in labs today. Identify the enzyme, and indicate the value of its being isolated from a thermophile. Suggest other reasons why enzymes from this bacterium (or other thermophiles) might also be valuable.

For selected answers, see Appendix A.

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